

KEITH, Australia

WMO: 958150

Lat: 36.1058S

Lon: 140.3272E

Elev: 28

StdP: 101.00

Time Zone: 9.50 (AUA)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.1	1.2	-3.5	2.8	24.2	-1.7	3.3	18.0	12.2	14.4	11.0	13.2	1.1	0	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	38.8	18.8	36.1	18.1	33.6	17.5	21.0	30.0	19.9	30.0	19.0	29.5	5.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.5	22.4	17.2	12.3	21.7	15.8	11.3	20.9	60.8	29.4	56.9	30.1	53.9	29.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.5	8.5	-2.1	43.8	0.8	1.8	-2.7	45.1	-3.2	46.2	-3.7	47.2	-4.2	48.6
			-2.4	22.7	0.8	1.1	-3.0	23.5	-3.4	24.1	-3.9	24.7	-4.5	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	21.2	20.7	18.8	15.6	12.6	10.1	9.8	10.0	11.4	14.0	17.0	19.4	(6)
	DBStd	5.54	4.64	4.37	4.34	3.58	2.83	2.31	2.14	2.44	2.96	4.22	4.61	4.56	(7)
	HDD10.0	116	0	0	0	1	7	27	31	30	14	6	1	0	(8)
	HDD18.3	1614	18	20	46	100	180	247	265	258	209	150	81	41	(9)
	CDD10.0	1944	347	300	272	167	87	30	24	30	56	129	212	290	(10)
	CDD18.3	402	106	87	61	17	1	0	0	0	1	15	42	72	(11)
	CDH23.3	6386	1678	1249	872	238	9	0	0	2	28	317	764	1228	(12)
	CDH26.7	3296	947	659	421	82	1	0	0	4	131	383	669		(13)
(14) Wind	WSAvg	4.0	4.3	4.1	3.8	3.5	3.7	3.5	4.2	4.0	4.0	4.1	4.2	4.4	(14)
(15) Precipitation	PrecAvg	469	20	18	21	32	50	56	57	61	53	44	32	26	(15)
	PrecMax	641	86	63	128	92	138	101	112	132	120	152	83	74	(16)
	PrecMin	249	2	0	0	1	8	4	17	14	11	6	3	2	(17)
	PrecStd	97	17	17	23	24	29	25	22	27	24	30	21	17	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	39.7	38.2	33.3	24.9	19.0	18.3	22.2	26.7	34.6	37.7	40.7	(19)
		MCWB	20.1	20.4	18.5	17.4	14.1	12.7	12.7	13.8	14.7	16.1	18.2	19.1	(20)
	2%	DB	39.6	36.6	34.4	28.7	21.8	16.9	16.4	19.1	23.1	30.3	34.5	37.3	(21)
		MCWB	19.4	18.9	18.0	15.4	14.0	12.3	11.8	12.5	13.6	15.2	16.9	17.8	(22)
	5%	DB	35.6	34.0	31.5	25.9	19.7	15.8	15.0	16.7	20.3	26.8	31.4	33.7	(23)
		MCWB	18.5	18.3	17.2	14.9	13.3	12.0	11.4	11.7	12.8	14.5	16.5	16.9	(24)
	10%	DB	31.9	30.9	28.2	23.2	17.9	14.6	14.0	15.0	18.0	23.5	28.0	30.3	(25)
		MCWB	17.5	17.5	16.4	14.2	13.0	11.4	11.0	11.2	12.3	13.7	15.6	16.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.4	20.5	18.5	16.4	14.9	13.7	14.7	16.2	17.8	20.7	21.6	(27)
		MCDB	29.8	34.8	27.4	24.9	20.0	16.2	16.0	19.3	22.7	25.7	28.1	30.3	(28)
	2%	WB	21.0	20.8	19.3	17.3	15.3	13.3	12.9	13.3	14.7	16.4	18.9	19.5	(29)
		MCDB	31.3	29.7	28.3	23.6	19.0	15.8	15.1	17.3	20.1	25.4	29.5	28.9	(30)
	5%	WB	19.7	19.6	18.5	16.2	14.4	12.5	12.1	12.4	13.7	15.4	17.7	18.3	(31)
		MCDB	31.9	29.3	28.1	23.1	17.9	15.0	14.3	15.7	18.8	24.1	27.7	29.6	(32)
	10%	WB	18.7	18.5	17.5	15.2	13.6	11.8	11.3	11.6	12.7	14.4	16.3	17.3	(33)
		MCDB	30.4	28.1	26.8	21.2	17.0	14.1	13.4	14.5	17.1	21.9	25.6	28.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	16.7	15.5	13.9	10.4	9.4	8.6	10.2	12.3	15.8	16.9	17.1	(35)
		MCDBR	22.8	20.7	19.3	17.6	12.5	10.3	9.4	12.6	16.6	21.1	21.8	22.8	(36)
	5% WB	MCWBR	7.3	6.9	6.9	7.6	6.5	6.5	6.1	7.5	8.7	8.4	7.8	7.6	(37)
		MCWBR	17.0	16.7	15.7	14.0	10.1	8.8	8.1	11.0	13.8	17.3	18.5	17.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.346	0.336	0.320	0.312	0.303	0.298	0.298	0.310	0.335	0.348	0.341	0.335	(40)
		taud	2.468	2.515	2.552	2.561	2.575	2.576	2.570	2.518	2.437	2.404	2.460	2.496	(41)
	Ebn at Noon		989	977	955	904	855	830	851	891	922	952	987	1004	(42)
		Edh at Noon	117	107	96	83	72	67	71	85	105	119	117	115	(43)
(44) All-Sky Solar Radiation	RadAvg		7.72	6.81	5.44	3.87	2.62	2.16	2.32	3.21	4.40	5.80	6.87	7.57	(44)
	RadStd		0.39	0.35	0.23	0.22	0.13	0.15	0.10	0.17	0.35	0.40	0.37	0.34	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	+0.44	N/A	N/A	+0.92	N/A	N/A	-16	-147	+168	+47

Nomenclature: See separate page